Signal and System Design (IGS2137)

Jump Together, Fly Farther!

School of Global Convergence Studies

인하대학교 국제학부

Week 3 Lab

ISE Department

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Content

- Introduction to MATLAB
- Creating Variables
- Some Useful MATLAB Functions
- Data Types
- Q & A
- Survey

SCRIPT FILES

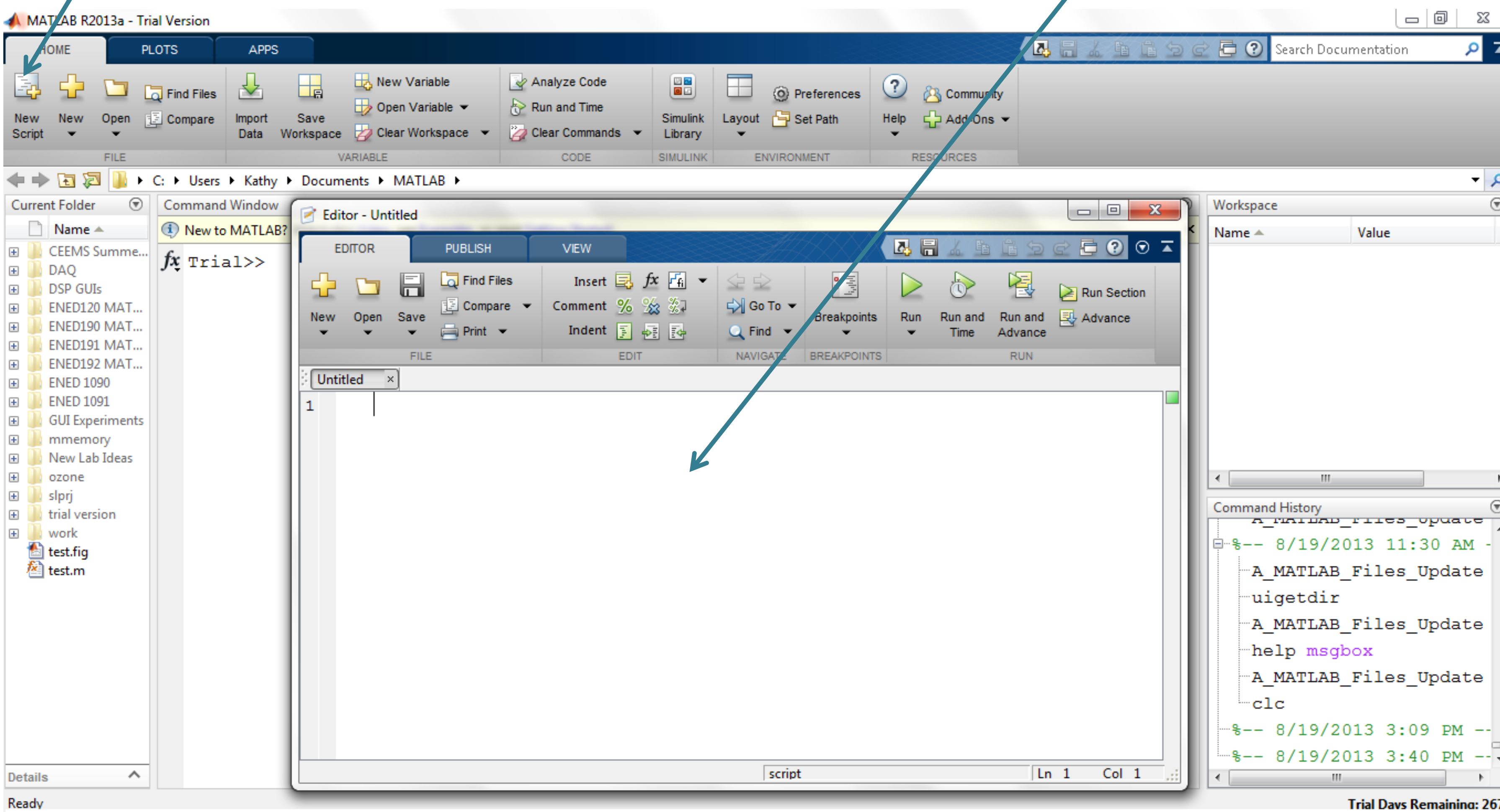
Script Files

- All of the pre-built commands that you use in MATLAB® are ***script files*** or ***functions*** (plot, mean, std, exp, cosd, …)
- MATLAB® allows the user to create his/her own customized m-files for specific applications or problems.
- A script file is simply a collection of executable MATLAB® commands. To create a new script file, click on the New Script icon on the left side of the Home Tab.

Script Files

Click on
New Script

Creates Blank
Script File



Script File: Procedure

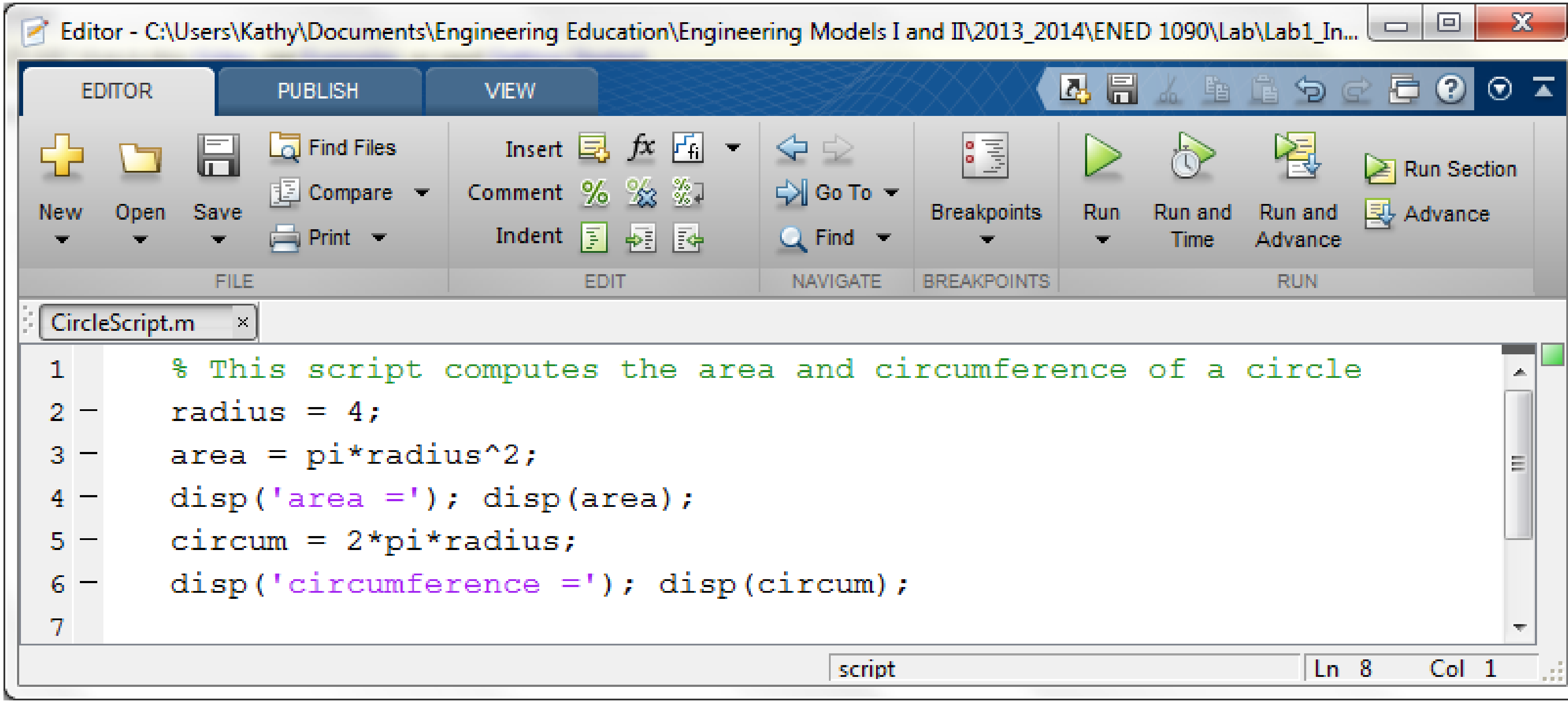
1. Type a set of executable commands in the editor window.
2. Save the file in an appropriate folder. ***When you pick a name for the file you must follow the same rules that MATLAB has for naming variables.***
3. Set the current directory in MATLAB® to the same place where you saved the script file.
4. To run the script file: Hit the Green Run Arrow in the Editor window or simply type the name of the file (without the .m extension) at the command prompt in the MATLAB command window.

The file must be in your current directory or it will not run!

Exercise 1: New Script File

- Right click in the current folder window in MATLAB and create a new folder named whatever you would like.
- Double click on the folder to make it your current folder.
- Clear your MATLAB workspace by typing `clear` at the command prompt.
- Click on New Script to open a blank script file.
- Type the commands on the next slide into the editor window then save the file as CircleScript in your newly created folder.

Exercise 1: Script File



Save the file as CircleScript in your newly created folder.

Note: Any line that starts with a % is a comment and turns green – it doesn't execute.

Exercise 1: Run the Script File

- Now run your script file by clicking on the Green Arrow in the m-file editor window.
- Notice that every single variable defined in the script file (radius, area, and circum) appears in the Workspace Wi ndow. Area and circum are also displayed in the Comma nd Window because of the **`disp`** command.
- Clear the workspace window by typing **`clear`** at the com mand prompt.
- At the command prompt, type the name of your script file : >> **`CircleScript`**. Note, that the results are exactly t he same as before.

Script Files

- A script file is simply a set of executable MATLAB commands saved together in a file. It behaves the same as entering each command sequentially in the command window. Every single variable defined appears in the workspace.
- Script files can be useful if you are going to execute a lot of M ATLAB commands. For example, suppose you execute 15 co mmands at the command prompt and discover an error in the first command that affected the last 14 commands. In the com mand window, you would have to fix the error in the first comm and then run the other 14 commands over again. If these com mands were in a script file, you could fix the first command an d re-run the script file – much faster !!

Exercise 2: Another Script File

- The equation for a 10 Hz sine wave with an amplitude of 3 is $3\sin(2\pi(10)t)$.
- A frequency of 10 Hz means the sine wave completes 10 cycles in 1 second.

Exercise 2: Another Script File

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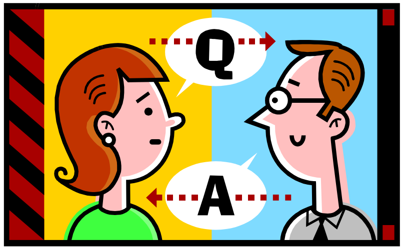
- The equation for a 10 Hz sine wave with an amplitude of 3 is $3\sin(2\pi(10)t)$.
- A frequency of 10 Hz means the sine wave completes 10 cycles in 1 second.

```
>> t = 0:0.02:0.6
>> y = 3*sin(2*pi*10*t)
>> plot(t,y)
```

Brief break (if on schedule)

Q&A?



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